11-28 第1篇

阅读理解（说明文） Why Do We Shiver When We Are Cold?

① Last Wednesday morning I stepped out of our warm bus shelter into a very cold wind. Within maybe **fifteen seconds**, a small, quick trembling ran across my shoulders and jaw. My teeth tapped together softly. I was **shivering**. The interesting thing was that I had not asked my body to do it; my body had simply decided. A classmate next to me, wearing a slightly thinner jacket, was shivering even harder. We both laughed at ourselves, half embarrassed and half amused. But behind this everyday little act is **a careful, ancient piece of biology** worth understanding. Why exactly do we shiver when we are cold, and when is it a good thing — or a warning?

② First, some quiet facts. The human body likes to keep its **core temperature** in a narrow safe band, around **36.5 ℃ to 37.5 ℃**. Inside this band, our enzymes — the small workers that power digestion, muscle movement, and thinking — work best. When the outside is cold, the body loses heat through the skin faster than it makes it. If nothing is done, the core begins to cool. Shivering is one of the body's **quickest, cheapest tools** to push that core back up toward the safe band.

③ How does shivering actually warm us? The trick is simple and surprising. Our **skeletal muscles** — the muscles you use to walk, write or lift a cup — can make heat not only when they do useful work, but also when they **contract without moving the body**. When the brain's small temperature centre, the **hypothalamus**, senses that the skin is too cold, it sends a fast signal down the nerves. Many small muscle fibres begin to contract and relax **very quickly and out of sync with each other**, at about **10 to 20 times per second**. The body does not go anywhere, but the muscles are burning sugar; and a side-product of that burning is **heat**. A good shiver can raise the body's heat production to about **5 times its normal resting level** for short periods. That is why, after a long cold wait, you suddenly feel a little warmer — the inner fire has been quietly turned up.

④ Why does the body use **small trembling**, rather than simply telling us to jump around? Because the body has limited time and energy. A big voluntary movement needs a plan and wastes energy on moving arms through space. A small shiver, almost without our permission, happens **automatically**, targets **the most muscles at once** (especially those in the shoulders, chest, and jaw), and produces **more heat per second per calorie**. Evolution has chosen a method that works even when you are **standing still, carrying a bag, or too cold to think clearly**. It is, in a sense, your body's quiet **backup heater**.

⑤ There is more than one step the body takes against cold. Before it shivers, it usually does **three smaller things** first, in this order: (1) **constricts blood vessels** in the skin and

limbs, to keep warm blood near the vital organs — this is why your fingers and nose go pale first; (2) **raises the tiny hairs** on your arms (goose-bumps), a feeble leftover from when our far ancestors had fur; (3) only if cooling continues, **triggers shivering**. The body, in short, only starts the "backup heater" when the gentle steps are not enough. Shivering is a sign that the body's cold defence has moved to **a higher level**.

⑥ Babies under three to six months **cannot shiver well**, because their small muscles are not yet strong enough. Their bodies use a different, beautiful method called **brown-fat thermogenesis**: a special fat stored in the neck and upper back burns itself directly to make heat. This is one reason why newborns need extra care in cold rooms. Interestingly, recent studies (2018, Harvard) show that adult humans also keep small amounts of brown fat in their upper back and chest, which activate during cold swims or long cold walks. In very cold swimmers, some heat comes from shivering, some from brown fat quietly helping in the background.

⑦ When is shivering **healthy**, and when is it a **warning**? Normal shivering in cold air usually stops within minutes after you warm up — put on a thick coat, go indoors, drink something warm, and the trembling fades. But if shivering continues for a long time despite warmth, or if it happens together with **pale lips, sleepiness, confusion, or slurred speech**, the body may be moving toward **hypothermia** (dangerous deep cooling) or **high fever** (the shiver here is the body trying to heat itself even higher to fight infection). Both cases need a trusted adult and a doctor's help. A shiver alone is an almost friendly sign; a shiver with those other signs is a small serious alarm.

⑧ A small practical side. Your shivering will calm much faster if you **move your body gently, cover your head, drink a warm liquid, and change out of wet clothes** — not wait. Many outdoor lessons on cold school trips now teach this sequence: "**Head, drink, move, dry.**" In that small order, your body's temperature control is helped, and the need to shiver ends. A smart student does not wait to be rescued by her shivers; she uses them as a reminder to take the four short steps.

⑨ So the next time you stand at a December bus stop and feel that small, fast trembling begin in your jaw, remember: your body has noticed the cold before your mind has. An ancient temperature centre in your brain has opened a quiet backup heater in your muscles, turning sugar into a soft, invisible flame. The shiver is not weakness. It is your body's small, clever way of taking care of you. **Listen to it; thank it; then, kindly, help it do less of its hard work** by putting on a hat, going inside, and warming your hands around a cup of tea.

注释： shiver 发抖；hypothalamus 下丘脑；constrict 收缩；hypothermia 低体温症；thermogenesis 产热

根据短文内容，选择最佳答案。

1. What is the normal safe range of the human body's core temperature, according to Paragraph ②?

A. 25 ˚C to 35 ˚C.
B. 36.5 ˚C to 37.5 ˚C.
C. 38 ˚C to 40 ˚C.
D. Any temperature.

2. How exactly does shivering produce heat?

A. By moving arms and legs wildly.
B. The hypothalamus sends a signal; many small skeletal-muscle fibres contract and relax very quickly and out of sync at about 10—20 times per second; the body does not go anywhere, but the muscles burn sugar, and a side-product of that burning is heat — raising heat production to about 5 times the resting level for short periods.
C. By increasing blood pressure only.
D. By sweating.

3. Before the body shivers, what smaller steps does it usually take first?

A. It sleeps and waits.
B. First, constricts blood vessels in the skin and limbs (fingers and nose go pale) to keep warm blood near vital organs; second, raises tiny hairs on the arms (goose-bumps); only if cooling continues does it trigger shivering — shivering is a sign that cold defence has moved to a higher level.
C. It removes clothes.
D. It jumps up and down.

4. What heat method do babies under 3—6 months use, and why?

A. They shiver more than adults.
B. They cannot shiver well because their small muscles are not yet strong enough, so their bodies use "brown-fat thermogenesis" — special fat stored in the neck and upper back burns directly to make heat; this is one reason newborns need extra care in cold rooms.
C. They make heat by crying.
D. They have no temperature control.

5. When should shivering be treated as a warning?

A. Whenever it lasts longer than a second.
B. Whenever it happens in a warm room.
C. When shivering continues for a long time despite warmth, or happens together with pale lips, sleepiness, confusion, or slurred speech — possible signs of hypothermia or high fever, both needing a trusted adult and a doctor.
D. When only the hands feel cold.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段安全区间
【解析】直引 36.5—37.5 °C。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第③段 10—20 Hz + 5 倍
【解析】产热机制直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】细节理解题
【定位】第⑤段三步顺序
【解析】血管收缩 → 鸡皮疙瘩 → 发抖直引。A、C、D 均错。故选 B。

4.
【答案】B
【题型】细节理解题
【定位】第⑥段褐色脂肪
【解析】直引。A、C、D 均错。故选 B。

5.
【答案】C
【题型】细节理解题
【定位】第⑦段警讯
【解析】持续 + 伴随症状直引。A、B、D 均错。故选 C。